week-6-lesson-1

exercise-1

Declare an integer array array_1 of size 10 and fill it with numbers from 0 to 9.
Print the elements in reverse order.
Find the sum of all elements in the array. Find index of maximum and minimum elements.
Print the results including index and values.

exercise-2

Declare an integer array array_2 of size 10 and fill it with numbers from -4 to 5.
Write a program which traverses the array and substitutes the negative values to -1 and positive numbers to 1.

exercise-3

Declare an integer array array_3 of size 10 and fill it with numbers from 1 to 10. Swap the 0th element and the last element, then swap the first element and the 9th element. Save the results in the same array and print the result.

exercise-4

Declare a two dimensional array array_4 of size 4 by 4 and fill it with numbers from 0 to 15.
Find the sum, maximum and minimum numbers in each row and in each column.
Find the sum, maximum and minimum numbers of the array.
Print the results including index and values.

exercise-5

Declare a two dimensional array array_5 of size 4 by 4 and fill it with numbers from 0 to 15. Print the array. Swap the elements of row 0 and row 2. Print the array. Swap the elements of column 0 and column 2. Print the array.

exercise-6

Declare a two dimensional array array_6 of size 4 by 4 and fill it with numbers from 1 to 16. Use variables for declaring the size of the array.
Print the array.
Print only the diagonal elements (from left top to right bottom) and print 0 for other elements.
Print only the diagonal elements (from right top to left bottom) and print 0 for other elements.

exercise-7

Declare a two dimensional array array_7 of size 5 by 5. Set all the elements to the right and above the diagonal to 1 and all the elements to the left and bottom the diagonal to -1. Set all the diagonal elements to 0.

exercise-8

Declare a two dimensional array array_8 of size 5 by 5. Set all the elements to the left and above the diagonal to 1 and all the elements to the right and bottom the diagonal to -1. Set all the diagonal elements to 0.

Declare and fill the matrix 4 by 5. Print the matrix. Transpose all the elements and save in a new matrix. Print the transposed matrix.